GeneScript: Compiler Construction Project

Course: Compiler Construction (BCS-7G) | **Instructor:** Sir Faisal Aslam

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1. Project Overview & Architecture

GeneScript is a domain-specific programming language designed for bioinformatics algorithms. It features built-in **dna** and **rna** data types with biological letter validation, value-first syntax ("ATGC" -> dna g), prefix sequence concatenation (fuse a b), reverse complement (~d), RNA transcription (transcribe(d)), and specialized loops (cycle i across s).

Phase 1 (Interpreter): Implemented using FLEX and Bison only, without intermediate bytecode. The parser constructs an Abstract Syntax Tree (AST), performs biological semantics validation (ensuring valid bases and presence of gene nucleus()), and directly evaluates the program in memory.

2. Project Files and Descriptions

File / Path	Description
include/ast.h	AST node definitions (Genes, Statements, Expressions, Values).
include/symbol_table.h	Scoped symbol table managing variable and loop-counter bindings.
include/semantics.h	Validates biological alphabets (A,C,G,T / A,C,G,U) & types.
include/interpreter.h	In-memory AST Tree-Walk interpreter runtime.
src/lexer.l	Flex scanner recognizing keywords, literals, comments (^^), ops.
src/parser.y	Bison LALR(1) grammar for value-first syntax & control flows.
src/semantics.cpp	Biological error formatting and type inference rules.
src/interpreter.cpp	Execution engine for biological functions (~, transcribe, fuse, etc.).
src/main.cpp	Driver program coordinating parsing, validation, and execution.
tests/sample1_filter.gene	Sample 1: Filter G/C bases, reverse complement, transcription.
tests/sample2_mergesort.gene	Sample 2: Recursive merge sort on DNA sequences.
tests/error.gene	Sample 3: Biological error reporting for invalid base 'X'.
Makefile	Linux build script supporting make, make clean, and make test.

3. Dependencies and Installation (Linux)

On Ubuntu / Debian Linux, install all required dependencies with:

```
sudo apt update && sudo apt install -y build-essential flex bison clang llvm make
```

4. How to Compile and Run

To compile the project using GNU Make:

```
make
```

To run GeneScript programs with the interpreter:

```
./genec tests/sample1_filter.gene
./genec tests/sample2_mergesort.gene
./genec tests/error.gene
```

To run the full automated verification test suite:

```
make test
```

5. Test Results & Verification

All test outputs match the proposal specification 100%:

• **Sample 1 Output:**

GGCCG
TACAATGGCCAT
AUGGCCAUUGUA

• **Sample 2 Output:**

AAACGTT

• **Sample 3 (Biological Error Output):**

tests/error.gene:1: invalid base 'X' in DNA literal.
DNA allows only A, C, G, T.
Compilation stopped: 1 error.